

DSA / Algorithms (Abdul Bari) - Day 16

45-day plan mapped to the Algorithms playlist

Strassen's Matrix Multiplication

Faster matrix multiplication via divide & conquer.

Watch these Abdul Bari videos

> 2.9 Strassens Matrix Multiplication

Key concepts

- Naive matrix multiply is $O(n^3)$.
- Strassen uses 7 multiplications instead of 8.
- Gives $\sim O(n^{2.81})$.

Key idea

Strassen reduces sub-multiplications with clever additions.

Tasks (1.5h)

- ☐ Watch the video
- ☐ Note the 7 products
- ☐ Compare with naive $O(n^3)$

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0